

Observed history
initial 64-d BPR

Panel + attribution
150 users
balanced labels

Matched random streams
choice and negative-selection draws

Six rounds
 $M = 240$; Top-20

Response
 δ fixed within pair

Within each round: score eligible candidates -> rank and display -> choice

Rank and choice
frozen model state

If accepted:
record outcome and remove item

No parameter update

Rank and choice
current model state

If accepted:
remove item and form update event

Online BPR update
3 SGD steps per accepted event

Round-6 paired contrast

$$\Delta_6(\delta) = D_6^{(C)}(\delta) - D_6^{(F)}(\delta)$$
$$\text{LMC}_6(\delta) = \Delta_6(\delta) - \Delta_6(0)$$

Frozen

Closed-loop